



Architecture principle

Semester should be the **connected student portal and action layer**, while the SIS remains authoritative for records, enrollment, holds, registration, and official degree data; the LMS remains authoritative for course delivery, submissions, and official grades unless Semester is explicitly deployed as the institution's LMS.

That distinction is critical: Semester can unify the experience without making unsafe claims or creating conflicting records. LTI Advantage, OneRoster, SAML/OIDC, SCIM, and institution-approved SIS gateways provide the interoperability foundation for this model. [¹⁸⁵⁴]^[1] ^[2]

```
Official systems
SIS / ERP / Registrar / LMS / Student Accounts / Financial Aid / Housing
    ↓ approved, minimum-field integrations
Semester operating layer
Identity · Path · Action Center · Search · Plan · Course Studio · Support · Career
    ↓
Student sees one connected journey and safe official handoffs
```

LMS roadmap

Stage 1: LMS connection and course context

Start by connecting to Canvas, Blackboard, Brightspace, Moodle, or another LMS through **LTI 1.3/LTI Advantage** and approved APIs.

Semester should receive only what it needs:

```
Course and section identity
Student's course membership
Instructor identity where authorized
Course launch context
Selected assignment metadata
Selected course materials
Approved deep links
Course AI policy
```

Semester should provide:

- Single sign-on launch from the LMS into Semester.
- Course workspace preloaded with the current course context.
- Deep links back to the official assignment, discussion, grade, or submission area.

- Source labels showing where course information originated.
- Last-sync timestamp and a fallback to the official LMS when data is unavailable.
- Student-selected materials for Study Studio.
- Faculty-approved source packs.
- Course-level AI policy controls.

LTI Advantage extends standard LTI with services including Names and Role Provisioning, Deep Linking, and Assignment and Grade Services. Its migration guidance notes that Assignment and Grade Services provide richer APIs for grade reporting than older LTI versions. [^1854] [2]

Stage 2: Semester Course Studio

Build a unified course workspace that connects planning, learning, support, and career context.

```
Course Overview
├─ Course source and freshness
├─ Syllabus and policies
├─ Assignments and timeline
├─ Materials and study library
├─ Study Studio
├─ Notes, files, and workspace
├─ Office hours and tutoring
├─ Course schedule and workload
├─ Course outcomes and skills
├─ Projects and portfolio evidence
├─ AI policy and permitted help
└─ Official LMS handoffs
```

This course context should follow a student everywhere:

Semester area	Course-aware behavior
Today	Shows upcoming course actions, deadlines, study prompts, and next class
My Path	Shows requirement fit and degree-plan impact
Plan	Shows meeting times, workload, study blocks, and conflicts
Study	Uses authorized materials for source-linked learning support
Workspace	Organizes notes, drafts, files, tables, and projects by course
Career	Connects confirmed coursework and projects to skills and portfolio evidence
Advising	Lets students choose course context for a meeting agenda
Search	Finds course materials, people, policies, tutoring, and assignments

Stage 3: assignments and learning workflow

Treat an assignment as a connected object, not merely an LMS row.

```
Assignment
→ Course
→ Learning outcomes
→ Due date
→ Rubric
→ Required sources
→ Student plan
→ Study support
→ Workspace draft
→ Human support
→ Official submission handoff
→ Feedback
→ Revision action
→ Student-approved portfolio evidence
```

Enhance assignment workflows with:

- Milestone planner for large projects.
- Rubric-to-checklist conversion.
- Student-selected source and citation workspace.
- Research and reading plan.
- Calendar and workload impact.
- Office-hour, tutoring, writing-center, and library links.
- Practice and teach-back mode.
- Revision checklist based on instructor feedback.
- Accessibility check before export.
- Final submission-preparation checklist.
- Official LMS submission handoff.
- Student-controlled portfolio saving after completion.

Stage 4: assessment and gradebook support

Start student-first. Build native learning evidence and progress tools before trying to replace the official gradebook.

Student experience

- Practice quizzes.
- Exam simulations.
- Confidence rating.
- Personal review queue.

- Source-linked explanations.
- Study plans tied to confirmed exam dates.
- Grade calculator using student-entered or approved inputs.
- “What score do I need?” scenarios with visible assumptions.
- Feedback interpreter and revision planner.
- Official-grade summary only where authorized and current.

Faculty experience

- Faculty-approved question banks.
- Learning-outcome tags.
- Rubric builder.
- Manual grading support.
- Feedback templates.
- Grade-release controls.
- Regrade and moderation workflow.
- Assignment AI-policy controls.
- Accessible assessment design.
- LMS-gradebook export or sync where institution-approved.

Official record boundaries

Semester can provide:

- Practice results
- Study progress
- Draft grade calculations
- Source-linked feedback
- Portfolio and skills evidence
- Student planning context

Official LMS/SIS retains:

- Final grades
- Official submissions
- Transcript records
- Enrollment status
- Grade changes
- Institutional academic record

OneRoster supports secure exchange of roster information, course materials, grades, and assessment results, and can be adopted in phased use cases rather than as a full replacement integration. ^[3] ^[1]

SIS sync roadmap

Stage 1: read-only student context

Use an institution-owned integration gateway or approved APIs—not direct production database access.

Minimum initial fields:

```
Active student status
Institution / school / program
Current term
Catalog course data
Course sections and meeting times
Enrollment status
Registration time window
High-level hold category
Advisor assignment if authorized
```

Avoid importing by default:

```
Detailed grades
Detailed financial records
Health data
Conduct records
Accommodation diagnoses
Counseling notes
Payment credentials
Full academic history beyond the active workflow
```

Stage 2: source and freshness model

Every imported item should show:

```
Source: Institution verified
System: Registrar / SIS / Catalog
Last updated: September 29, 2026 at 10:15 AM
Freshness: Current / Stale / Unavailable
Official fallback: Open registration portal
```

Use these standard states everywhere:

State	Meaning
Institution verified	Received from an approved, current institutional source
Imported	Received from a connected non-authoritative or user-authorized source
Student entered	Entered or confirmed by the student
Estimated	Calculated by Semester; not official

State	Meaning
Needs review	Extracted or incomplete; requires confirmation
Stale	Source has not refreshed within the expected window
Unavailable	Source could not be reached or data is not authorized

Stage 3: event-driven synchronization

Use a durable integration architecture:

```
SIS/LMS event or scheduled sync
→ signature and schema validation
→ durable inbox
→ idempotency and replay check
→ field mapping
→ source precedence check
→ domain event
→ Semester update
→ Action Center update if relevant
→ audit event
→ reconciliation job
```

Rules:

- Use webhook signature verification.
- Use timestamp/replay protection.
- Use durable queues.
- Use idempotency keys.
- Do not create duplicate actions on repeated events.
- Do not overwrite student-created notes or plans with external data.
- Show stale data rather than silently presenting it as current.
- Send ambiguous failures to an exception queue.

Student portal blueprint

Keep the permanent student navigation simple:

```
Today
My Path
Search
Plan
Me
```

Everything else opens contextually.

Today

Path Snapshot
Most Important Action
Next 3-5 Actions
Today's schedule
Upcoming deadlines
Study prompt
Registration status
Campus and career context
Source freshness
Quick actions

It answers:

Am I on track?
What matters now?
What should I do next?
Who can help?

My Path

Degree / credential map
Requirement detail
Term plan
Registration Center
Scenario comparison
Transfer planning
Study abroad impact
Graduation estimate
Advisor agenda

Search

Ask Semester
Courses
People
Campus services
Policies
Events
Opportunities
Study materials
Portfolio evidence
Career resources

Search should answer and act—not only return links.

Plan

Weekly schedule
Term plan
Study blocks
Assignments
Meetings
Work and travel constraints
Workload forecast
Deadlines
Calendar connections

Me

Study Library
Workspace
Career and Portfolio
Community and Connections
Membership and Billing
Connected Accounts
Privacy and Sharing
Accessibility Preferences
AI Controls
Settings

Academic advising and support services

Advisor Meeting Mode

Create a controlled shared workspace that begins with the student—not the advisor.

Student selects what to share
→ Semester creates agenda
→ Advisor sees only approved plan/context
→ Meeting produces student-reviewed follow-up actions
→ Student can revoke future access

Advisor view can include:

- Student-selected goal.
- Current path summary.
- Open requirements.
- Chosen scenarios.
- Course options and schedule conflicts.
- Student questions.
- Relevant deadlines.

- Meeting agenda.
- Follow-up actions.
- Shared documents only when selected.

Advisor view must not automatically include:

- Private notes.
- Health or counseling context.
- Financial details.
- Study behavior.
- Full calendar.
- AI history.
- Supporter information.
- Sensitive institutional records.

Support routing

Build a single support-intelligence flow:

```
Student asks for help
→ Semester identifies possible resource category
→ Shows verified offices and why they may help
→ Displays source, hours, accessibility, and official contact path
→ Helps prepare questions
→ Creates student-approved follow-up action
→ Tracks only what the student chooses to track
```

Support services can include:

- Academic advising.
- Tutoring.
- Writing center.
- Library and research support.
- Accessibility services.
- Financial aid.
- Student accounts.
- Housing and residence life.
- International student services.
- Veteran services.
- Career center.
- Counseling and wellness resources.

- Food, transportation, emergency aid, and other basic-needs services.
- Study abroad.
- Student organizations and peer mentoring.

For emergency, health, legal, or crisis situations, Semester should direct users to official urgent resources according to institutional policy. It should not claim to be emergency response or attempt clinical triage.

Holds and administrative blocks

Hold architecture

Holds must be treated as **source-controlled administrative states**, not generic tasks.

```
Official system detects hold
→ Semester receives only approved high-level category and status
→ Semester shows neutral explanation
→ Semester shows relevant official resolution route
→ Student can prepare questions/checklist
→ Action Center tracks student-controlled follow-up
→ Official system remains authoritative for release
```

What Semester should show

```
Status: Action required before registration
Category: Advising / Financial / Records / Immunization / Library / Other
Source: Institution verified
Updated: [timestamp]
What this may affect: Registration may be unavailable until resolved
Next step: Visit the official resolution page or contact the responsible office
```

What Semester should not show by default

```
Detailed financial balances
Bank or payment information
Medical/immunization details
Disciplinary details
Confidential case notes
Sensitive legal documentation
Internal staff-only comments
Unverified cause or outcome
```

Hold workflow

Hold category	Semester supports	Official system retains
Advising	Checklist, advisor questions, appointment handoff	PIN release, approval, official clearance
Financial	High-level deadline, payment-plan link, aid-office questions	Balance ledger, payment, aid decision, release
Records	Document checklist, registrar contact, official link	Record review and hold release
Immunization / health	Neutral status and health-services handoff	Health data, documentation, clearance
Library	Resource link and contact	Fine/account detail and release
International	Compliance reminder, official adviser handoff	Visa/documentation record and decision
Conduct	General official contact path only	Case information, adjudication, release
Unknown	“Contact the official office” fallback	Internal determination

Hold design rules

- Use non-stigmatizing language.
- Never call a student “blocked,” “noncompliant,” “at risk,” or “failing.”
- Always show source and timestamp.
- Never promise the hold will be removed.
- Do not infer the reason from unrelated data.
- Provide a “This looks wrong” support route.
- Allow students to mark their own follow-up as done, but do not mark the official hold resolved until verified source data confirms it.
- Keep restricted details out of notifications and lock-screen previews.
- Require fresh data before high-stakes registration guidance.

FERPA and institutional control

When Semester processes institutional education records, the institution must determine the legal basis and ensure appropriate contractual and governance controls. Under FERPA’s school-official exception, an outside provider must perform an institutional service or function, operate under the institution’s direct control regarding the use and maintenance of education records, use PII only for the authorized purpose, avoid unauthorized redisclosure, and meet the institution’s legitimate-educational-interest criteria. [\[4\]](#) [\[5\]](#)

Operationally, Semester should use:

Written agreement / DPA
Institution-defined authorized purpose
Minimum necessary fields
Direct institutional control

No unauthorized redisclosure
No public-model training without explicit authorization
Tenant isolation
Role and object-level authorization
Retention and deletion schedule
Source lineage
Audit logs
Incident response
Institution-controlled configuration

Recommended build order

First: student clarity and registration

1. Today dashboard and Action Center.
2. Path Snapshot.
3. Term planner and schedule conflicts.
4. Registration Center.
5. Primary and backup course planning.
6. Advisor Meeting Mode.
7. High-level hold cards with official handoff.
8. Source labels and freshness.

Second: course and learning depth

1. LTI launch and course context.
2. Syllabus intelligence with review.
3. Course Studio.
4. Assignment object and workspace.
5. Source-linked Study Studio.
6. Faculty-approved source packs.
7. Course AI policy.
8. Practice assessment and student grade calculator.

Third: support and institution maturity

1. Campus Graph and verified resource directory.
2. Support routing.
3. Office action publisher.
4. Content governance.
5. Integration health console.

6. SIS read-only gateway.
7. OneRoster and LTI expansion.
8. Role, consent, and audit console.

Fourth: strategic differentiation

1. Transfer and dual-enrollment center.
2. Scenario and graduation simulator.
3. CASE-compatible skills and competency graph.
4. Portfolio and verified skills passport.
5. Group-project workspace.
6. Accessibility Passport.
7. Study-abroad and athlete planning.
8. Campus community and opportunity network.

Final test

Semester's LMS, registration, advising, and support architecture is strong when a student can receive a confusing signal—a hold, course conflict, assignment deadline, registration window, advisor instruction, or support need—and immediately understand:

1. What it means.
2. Where the information came from.
3. Whether it is current and official.
4. What they can safely do next.
5. Who can help.
6. What they need to do in an official external system.
7. What Semester will remember, share, or let them revoke.

That is how Semester becomes the trusted student portal and connected university operating layer—without overreaching into institutional authority.



1. <https://www.1edtech.org/standards/oneroster>
2. <https://www.imsglobal.org/lti-advantage-faq>
3. <https://www.1edtech.org/blog/rostering-resources-and-gradebook-standards>
4. <https://studentprivacy.ed.gov/faq/who-school-official-under-ferpa>
5. https://studentprivacy.ed.gov/sites/default/files/resource_document/file/Vendor_FAQ.pdf
6. <https://www.1edtech.org/standards/lti/suggested-lti-advantage-requirements-rfps>
7. <https://www.imsglobal.org/lti-advantage-overview>

8. <https://www.1edtech.org/standards/oneroster/OneRoster-Procurement-Guide>
9. https://standards.1edtech.org/lti/guides/implementation_guide/implementation-guide
10. <https://standards.1edtech.org/oneroster/specifications/standards/v1p2>
11. <https://learn.1edtech.org/mod/glossary/view.php?id=154&mode&hook=ALL&sortkey&sortorder&fullsearch=0&page=10>
12. <https://standards.1edtech.org/lti/guides/migration/migration-guide>
13. <https://blog.promise.legal/edtech-school-district-vendor-agreement/>
14. <https://www.deepinspect.ai/blog/edtech-ferpa-ai-compliance>
15. <https://www.1edtech.org/standards/lti>